A NOTE ON BUKH'S CONSTRUCTION OF EXTREMAL GRAPHS WITHOUT SMALL BICLIQUES

HENG LI AND XIZHI LIU

Abstract. We prove that for every fixed

$$c > 3 \cdot 4^{2/3} = 7.559526299\ldots,$$

and every sufficiently large s, one has

$$\mathrm{ex}(n, K_{s,\lfloor c^s \rfloor}) = \Omega_s(n^{2-1/s})$$

for all sufficiently large n. This improves the exponential constant in Bukh's random algebraic construction of extremal graphs without small bicliques. The key new ingredient is a quartic auxiliary sieve: we prove a sharp span bound for cubic circuits and use it to construct s-wise cubic-independent complete intersections of comparatively small degree. In quantitative form, the construction gives the threshold

$$t > 3^s 4^{(2/3+\varepsilon)s} s^{O(s^{2/3})}$$

for every fixed $\varepsilon > 0$. This note is not intended for publication.

1. Introduction

For a graph F, let $\mathrm{ex}(n, F)$ be the maximum number of edges in an n-vertex graph containing no copy of F. The Kővari–Sós–Turán theorem gives

$$\mathrm{ex}(n, K_{s,t}) = O_{s,t}(n^{2-1/s}).$$

The difficult direction is to construct $K_{s,t}$-free graphs with $\Omega_s(n^{2-1/s})$ edges while keeping t as small as possible as a function of s.

Bukh [2] constructed such graphs for

$$t > 9^s s^{4s^{2/3}}.$$

The factor 9^s comes from the use of a cubic edge polynomial together with an auxiliary variety cut out by roughly s cubic equations. The present note keeps Bukh's cubic edge polynomial, but replaces the auxiliary cubic sieve by a quartic sieve. The point is that a minimal cubic dependence imposes all its point conditions on quartics, so quartic equations are more efficient for killing cubic circuits.

The new geometric input is the following theorem.

Theorem 1.1 (Cubic Veronese circuit span bound)**.** *Let k be a field, and let $P \subset \mathbb{P}^d(k)$ be a finite set of distinct k-rational points which spans $\mathbb{P}^d$. Suppose that P is minimally dependent on cubic forms, i.e.*

$$H_P(3) = |P| - 1$$

and every proper subset $P' \subsetneq P$ satisfies $H_{P'}(3) = |P'|$. Then

$$|P| \geq 3d + 2.$$

The bound is sharp. If P consists of $3d+2$ points on a rational normal curve $C \subset \mathbb{P}^d$, then cubics restrict to $\mathcal{O}_{\mathbb{P}^1}(3d)$ on C, whose space of sections has dimension $3d+1$. Thus $3d+2$ general points on C form a cubic circuit.

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Theorem 1.1 is proved in Section 3. The proof uses a linear Kneser inequality in the split function algebra k^P. Section 5 applies the span bound to a quartic independence sieve. Section 6 records the form of Bukh's random algebraic construction needed here. Section 7 proves the extremal graph statement.

2. Hilbert functions and cubic circuits

Let $P \subset \mathbb{P}^b(k)$ be a finite set of distinct k-rational points over a field k. Its Hilbert function is

$$H_P(a) = \dim_k \operatorname{im}\big(H^0(\mathbb{P}^b, \mathcal{O}_{\mathbb{P}^b}(a)) \to k^P\big).$$

Equivalently, $H_P(a)$ is the rank of the evaluation map of degree-a homogeneous forms on P.

Definition 2.1. A finite point set $P \subset \mathbb{P}^b(k)$ is *minimally cubic-dependent*, or a *cubic circuit*, if

$$H_P(3) = |P| - 1$$

and every proper subset $P' \subsetneq P$ has

$$H_{P'}(3) = |P'|.$$

We shall use the following elementary Hilbert-function fact.

Lemma 2.2 (Quartics separate a cubic circuit)**.** *If P is a cubic circuit of size u, then*

$$H_P(4) = u.$$

Proof. The Hilbert function of a finite set is nondecreasing and eventually equal to u. Since $H_P(3) = u - 1$, the only way $H_P(4) < u$ could occur is if $H_P(4) = u - 1$.

After a harmless field extension, choose a linear form ℓ not vanishing at any point of P. Let

$$R(P) = \bigoplus_{d \geq 0} R_d(P)$$

be the homogeneous coordinate algebra of P, so $\dim R_d(P) = H_P(d)$. Multiplication by ℓ induces injective maps

$$R_d(P) \xrightarrow{\cdot \ell} R_{d+1}(P)$$

between all degree pieces, since ℓ is nonzero at every point of P. If $\dim R_3(P) = \dim R_4(P)$, this multiplication map is an isomorphism from degree 3 to degree 4, so $R_4(P) = \ell R_3(P)$. The algebra $R(P)$ is standard graded, hence $R_{d+1}(P) = R_1(P)R_d(P)$ for all d. Therefore

$$R_5(P) = R_1(P)R_4(P) = R_1(P)\ell R_3(P) = \ell R_4(P),$$

and the same induction gives $R_{d+1}(P) = \ell R_d(P)$ for every $d \geq 4$. Since multiplication by ℓ is injective, all these maps are isomorphisms. Hence $\dim R_d(P) = u - 1$ for all $d \geq 3$, contradicting the fact that the Hilbert function of a finite set eventually reaches u. Therefore $H_P(4) = u$. □

3. A linear Kneser input and the cubic circuit span bound

The proof of Theorem 1.1 uses a standard linear Kneser inequality. We state it only in the split function-algebra form needed below.

Theorem 3.1 (Linear Kneser inequality for split function algebras)**.** *Let k be an infinite field, let Ω be a finite set, and let*

$$A = k^\Omega$$

with pointwise multiplication. Let $X, Y \subset A$ be finite-dimensional k-subspaces containing the constant function 1*. Put*

$$XY = \operatorname{span}_k\{xy : x \in X,\ y \in Y\}$$

and let

$$H = \{h \in A : hXY \subset XY\}$$

be the stabilizer algebra of XY. Then

$$\dim XY \geq \dim X + \dim Y - \dim H.$$

Remark 3.2. This is the Kneser-Diderrich inequality for finite-dimensional associative algebras, applied to the split commutative algebra k^Ω. It follows from Beck–Lecouvey's theorem [1, Theorem 4.1]. We use only this split case, where subalgebras of k^Ω are the algebras of functions constant on the blocks of finite partitions of Ω. The hypotheses of Beck–Lecouvey's theorem are satisfied here: over an infinite field, k^Ω has only finitely many subalgebras, and every subspace containing 1 contains invertible elements. Since the dimension inequalities used below are invariant under base-field extension, this form also applies after extending an arbitrary field to an infinite one.

Lemma 3.3 (Idempotents in split subalgebras)**.** *Let k be infinite and let $H \subset k^\Omega$ be a subalgebra containing the constants. If $\dim H > 1$, then H contains the indicator function 1_Q of a nonempty proper subset $Q \subsetneq \Omega$.*

Proof. Choose a nonconstant function $h \in H$. Its finite image consists of at least two values, say $\alpha_1, \ldots, \alpha_m \in k$. Choose one value, say α_1, and let $p(T) \in k[T]$ be a polynomial with $p(\alpha_1) = 1$ and $p(\alpha_i) = 0$ for $i > 1$. Then $p(h) \in H$ is exactly the indicator function of the nonempty proper fiber $h^{-1}(\alpha_1)$. ☐

We now prove the span bound.

Theorem 3.4 (Cubic circuit span bound)**.** *Let $P \subset \mathbb{P}^d(k)$ be a finite set of distinct k-rational points spanning $\mathbb{P}^d$. If P is a cubic circuit, then*

$$|P| \geq 3d + 2.$$

Proof. All relevant ranks and the minimal-dependence property are unchanged by extending the base field. We may therefore assume that k is infinite.

Let

$$N = |P|.$$

Choose a linear form ℓ nonzero at every point of P and dehomogenize by the affine chart $\ell = 1$. Let

$$A \subset k^P$$

be the vector space of restrictions to P of affine linear functions. Since P spans $\mathbb{P}^d$,

$$\dim A = d + 1.$$

Let

$$A^2 = \operatorname{span}\{fg : f, g \in A\}, \qquad A^3 = A \cdot A^2.$$

The spaces A^i are exactly the restrictions to P of polynomials of affine degree at most i, equivalently the dehomogenized restrictions of homogeneous forms of degree i. Hence

$$\dim A^i = H_P(i).$$

Since P is a cubic circuit,

$$\dim A^3 = N - 1.$$

Thus there is a unique, up to scalar, nonzero linear functional

$$\lambda \in (k^P)^*$$

annihilating A^3. Write

$$\lambda(f) = \sum_{p \in P} \lambda_p f(p).$$

Minimality of the cubic dependence implies

$$\lambda_p \neq 0 \qquad \text{for every } p \in P.$$

Indeed, if some $\lambda_p = 0$, then the support of λ would be contained in a proper subset of P, giving a cubic dependence on that proper subset, contrary to minimality.

We first prove an upper bound for $\dim A^2$. For every $q \in A^2$ and every $a \in A$,

$$aq \in A^3,$$

so

$$0 = \lambda(aq) = \sum_{p \in P} \lambda_p q(p) a(p).$$

Therefore the vector

$$(\lambda_p q(p))_{p \in P} \in k^P$$

lies in the annihilator of A. Since $A \hookrightarrow k^P$ is injective of dimension $d + 1$, this annihilator has dimension

$$N - (d + 1) = N - d - 1.$$

Because all λ_p are nonzero, the map

$$q \longmapsto (\lambda_p q(p))_{p \in P}$$

is injective on A^2. Hence

$$\dim A^2 \le N - d - 1. \tag{1}$$

We next prove a lower bound. Let

$$H = \{h \in k^P : hA^2 \subset A^2\}$$

be the stabilizer algebra of A^2. Applying Theorem 3.1 with $X = Y = A$ gives

$$\dim A^2 \ge 2 \dim A - \dim H = 2d + 2 - \dim H. \tag{2}$$

We claim that

$$\dim H = 1.$$

Suppose instead that $\dim H > 1$. By Lemma 3.3, H contains the indicator function $e = 1_Q$ of a nonempty proper subset

$$Q \subsetneq P.$$

Because $e \in H$, we have $eA^2 \subset A^2$. Consequently,

$$eA^3 = e(A \cdot A^2) = A \cdot (eA^2) \subset A \cdot A^2 = A^3.$$

Since λ annihilates A^3, for every $f \in A^3$ we have

$$0 = \lambda(ef) = \sum_{p \in Q} \lambda_p f(p).$$

The coefficients λ_p are nonzero on Q, so this gives a nontrivial cubic dependence on the proper subset Q. This contradicts the minimality of P. Hence $\dim H = 1$.

Substituting into (2),

$$\dim A^2 \ge 2d + 1. \tag{3}$$

Combining (1) and (3),

$$2d + 1 \le N - d - 1.$$

Therefore

$$N \ge 3d + 2.$$

□

4. Dimension bounds for cubic circuits

For integers $u, b \geq 1$, let $\Phi_u(b,3)$ denote the parameter space of ordered u-tuples of distinct points in $\mathbb{P}^b$ which form a cubic circuit. Let

$$\varphi_u(b,3) = \dim \Phi_u(b,3).$$

We need an upper bound for $\varphi_u(b,3)$.

The following estimate is the fixed-span part of Bukh's dependence count.

Lemma 4.1 (Bukh's fixed-span estimate)**.** *Let $\Phi'_u(d,3)$ be the family of cubic circuits of size u spanning a fixed projective d-space. Then*

$$\dim \Phi'_u(d,3) \leq (u-d-1)(d+1).$$

Remark 4.2. This is Bukh's tangent-space estimate for minimally dependent configurations, specialized to $m = 3$; see [2, Lemma 21]. It is one of the inputs from Bukh's paper that we use without reproving.

Proposition 4.3 (Improved cubic-circuit dimension bound)**.** *For all u and b with $u \leq b$,*

$$\varphi_u(b,3) \leq \frac{u+1}{3}\left(b + \frac{u+1}{3}\right).$$

Proof. Let a cubic circuit P of size u span a projective d-space. By Theorem 3.4,

$$d \leq \frac{u-2}{3}.$$

For fixed d, the choice of the d-plane in $\mathbb{P}^b$ contributes

$$\dim \mathrm{Gr}(d,b) = (d+1)(b-d)$$

parameters, and Lemma 4.1 contributes at most

$$(u-d-1)(d+1)$$

more. Hence

$$\varphi_u(b,3) \leq \max_{0 \leq d \leq (u-2)/3} (d+1)(b-d) + (u-d-1)(d+1).$$

The expression is

$$(d+1)(b+u-2d-1).$$

For $u \leq b$ this expression is increasing on $0 \leq d \leq (u-2)/3$, since its derivative is

$$b + u - 4d - 3 \geq b + u - \frac{4(u-2)}{3} - 3 = b - \frac{u}{3} - \frac{1}{3} > 0.$$

Thus the maximum over the allowed integer values of d is at most the value of the same increasing real function at the real endpoint $d = (u-2)/3$. This gives

$$\varphi_u(b,3) \leq \frac{u+1}{3}\left(b + \frac{u+1}{3}\right).$$

□

5. The quartic sieve

We now construct an auxiliary variety which is s-wise cubic-independent using only quartic equations.

Definition 5.1. A variety $W \subset \mathbb{P}^b$ is *s-wise cubic-independent* if every subset of $W(\bar{k})$ of size at most s imposes independent conditions on cubic forms, i.e. contains no cubic circuit of size at most s.

Proposition 5.2 (Quartic independence sieve)**.** *Fix $\varepsilon > 0$. Let*

$$Z = \left\lceil \left(\frac{2}{3} + \varepsilon\right) s \right\rceil,$$

and let

$$\rho = O(s^{2/3})$$

be any integer satisfying

$$\binom{\rho + 4}{3} \geq s^2.$$

Set

$$b = s + \rho + Z.$$

For all sufficiently large s, a general complete intersection of Z quartic hypersurfaces in $\mathbb{P}^b$ is s-wise cubic-independent.

Proof. Let $5 \leq u \leq s$. Cubic circuits have size at least 5, since four points always impose independent conditions on cubics. For a cubic circuit P of size u, Lemma 2.2 gives

$$H_P(4) = u.$$

Thus the condition that one quartic form vanish on P imposes u independent linear conditions on the space of quartic forms. Therefore Z quartic equations impose Zu independent conditions on P.

By Proposition 4.3, the dimension of the family of cubic circuits of size u in $\mathbb{P}^b$ is at most

$$\varphi_u(b, 3) \leq \frac{u+1}{3}\left(b + \frac{u+1}{3}\right).$$

It suffices to verify

$$Zu > \frac{u+1}{3}\left(b + \frac{u+1}{3}\right) \tag{4}$$

for all $5 \leq u \leq s$.

Since

$$Z = \left(\frac{2}{3} + \varepsilon\right) s + O(1),$$

and

$$b = s + \rho + Z = \left(\frac{5}{3} + \varepsilon\right) s + O(s^{2/3}),$$

the difference between the two sides of (4) is at least

$$\begin{aligned} &\left(\frac{2}{3} + \varepsilon\right) su - \frac{u+1}{3}\left(\left(\frac{5}{3} + \varepsilon\right) s + \frac{u+1}{3} + O(s^{2/3})\right) \\ &\quad = s\left(\frac{u-5}{9} + \varepsilon\frac{2u-1}{3}\right) - \frac{(u+1)^2}{9} - O(us^{2/3}). \end{aligned}$$

For $u = 5$, this is

$$3\varepsilon s - O(s^{2/3}) > 0$$

for all sufficiently large s.

Now let $6 \le u \le s$. If $u \le s^{2/3}$, then $(u+1)^2 = O(us^{2/3})$, while the positive term $\varepsilon(2u-1)s/3$ is $\gg_\varepsilon us$, and hence dominates the error. If $u > s^{2/3}$, write $u = \lambda s$ with $s^{-1/3} < \lambda \le 1$. The leading contribution from the non-error terms is

$$s^2\left(\frac{\lambda(1-\lambda)}{9} + \frac{2\varepsilon\lambda}{3}\right) + O(s),$$

which is at least $c_\varepsilon \lambda s^2$ for some $c_\varepsilon > 0$. This dominates the error term $O(us^{2/3}) = O(\lambda s^{5/3})$. Thus (4) holds uniformly for $5 \le u \le s$ once s is large.

Let $\mathcal{P}$ be the affine parameter space of ordered Z-tuples of quartic forms. For each u, the incidence variety of pairs consisting of a cubic circuit P of size u and a Z-tuple of quartics all vanishing on P has dimension strictly smaller than $\dim \mathcal{P}$. Its projection to $\mathcal{P}$ is constructible, and its Zariski closure is therefore a proper closed subset of $\mathcal{P}$. Since there are only finitely many $u \le s$, the union of these closures is still a proper closed subset. Outside this closed subset, the common zero locus contains no cubic circuit of size at most s.

Intersecting the resulting nonempty open set with the standard nonempty open locus of Z-tuples cutting out a complete intersection of codimension Z gives the claimed general complete intersection. Hence a general complete intersection is s-wise cubic-independent. □

Corollary 5.3 (Finite-field specialization)**.** *For every fixed $\varepsilon > 0$ and all sufficiently large s, the following holds over all sufficiently large finite fields of characteristic > 4. There exists a variety*

$$W \subset \mathbb{P}^b_{\mathbb{F}_q}$$

which is a smooth complete intersection of

$$Z = \left\lceil \left(\frac{2}{3} + \varepsilon\right) s \right\rceil$$

quartic hypersurfaces, has dimension $s + \rho$, degree at most 4^Z, has

$$|W(\mathbb{F}_q)| \gg_s q^{s+\rho},$$

and is s-wise cubic-independent over $\overline{\mathbb{F}}_q$.

Proof. Let $\mathcal{P}$ be the affine parameter space of ordered Z-tuples of quartic forms in $b+1$ variables. The incidence argument in Proposition 5.2 shows that the locus of tuples whose common zero locus contains a cubic circuit of size at most s is contained in a proper Zariski-closed subset of $\mathcal{P}_{\overline{\mathbb{F}}_q}$. This closed subset is defined over $\mathbb{F}_q$, since the incidence construction is defined over $\mathbb{F}_q$. Its complement is a nonempty open condition guaranteeing s-wise cubic-independence over the algebraic closure.

Independently, the locus where the common zero locus is a smooth complete intersection of codimension Z is a nonempty Zariski-open subset in characteristic > 4 by the usual Bertini theorem, since $b - Z = s + \rho \ge 1$. The intersection of these two nonempty open conditions is a nonempty open subset $\mathcal{U} \subset \mathcal{P}$, defined over $\mathbb{F}_q$. A nonempty open subset of affine space over $\mathbb{F}_q$ has an $\mathbb{F}_q$-rational point for all sufficiently large q; the required lower bound on q depends only on s and ε. Choose the corresponding quartics.

Their common zero locus W is a smooth complete intersection in $\mathbb{P}^b$ of codimension Z. Therefore

$$\dim W = b - Z = s + \rho$$

and

$$\deg W = 4^Z.$$

For large s we have $s+\rho \ge 2$; by the Lefschetz connectedness theorem, a smooth complete intersection of dimension at least 2 in projective space is geometrically connected, and hence geometrically irreducible. Lang–Weil gives

$$|W(\mathbb{F}_q)| = q^{s+\rho} + O_s(q^{s+\rho-1/2}),$$

so $|W(\mathbb{F}_q)| \gg_s q^{s+\rho}$ for all sufficiently large q. The chosen tuple avoids the closed bad locus, so $W(\overline{\mathbb{F}}_q)$ contains no cubic circuit of size at most s. $\square$

6. Interface with Bukh's random algebraic construction

We use the following form of Bukh's random algebraic construction. It is a direct variant of [2, Lemma 23], with the degree of the auxiliary variety left as a parameter. Following Bukh, for integers $i, T \geq 1$ write

$$M_i(T) = \min\left\{m \geq 0 : \binom{m+i}{i} \geq T\right\}.$$

Proposition 6.1 (Bukh interface with an auxiliary variety of degree Δ)**.** *Let $s \geq 2$ and let ρ satisfy*

$$\binom{\rho+4}{3} \geq s^2.$$

Suppose that, for every sufficiently large prime q, there is a pure-dimensional s-wise cubic-independent variety

$$W \subset \mathbb{P}^b_{\mathbb{F}_q}$$

of dimension $s+\rho$, degree at most Δ, and with $\gg_s q^{s+\rho}$ rational points. Then Bukh's cubic random algebraic graph construction gives, for all sufficiently large n, graphs with

$$\Omega_s(n^{2-1/s})$$

edges and no $K_{s,t}$ whenever

$$t > 3^s \Delta \prod_{i=1}^{\rho} M_i(s^2),$$

where $M_i(T)$ is defined above. Moreover Bukh's choice of $\rho = O(s^{2/3})$ gives

$$\prod_{i=1}^{\rho} M_i(s^2) = s^{O(s^{2/3})}.$$

Proof. We recall the part of Bukh's proof in which the degree of the auxiliary variety enters. Run the construction of [2, Lemma 23] with the present pure-dimensional variety W in place of Bukh's auxiliary complete intersection. The slicing step uses hypersurfaces of degrees $M_i(s^2)$, $1 \leq i \leq \rho$, on each side. Since $|W(\mathbb{F}_q)| \gg_s q^{s+\rho}$ and $\dim W = s+\rho$, Bukh's point-counting argument gives two sliced sets $L, R \subset W(\mathbb{F}_q)$ with $|L|, |R| \asymp_s q^s$. Join $x \in L$ to $y \in R$ when a random bidegree-$(3,3)$ polynomial vanishes at (x, y). The expected number of edges is $\Omega_s(q^{2s-1})$.

The only change from [2, Lemma 23] is the Bezout factor used to bound common neighborhoods. Fix s vertices on one side. Because W is s-wise cubic-independent, freezing those vertices in the bidegree-$(3,3)$ edge polynomial gives s independent cubic equations on the other side. The common neighborhood is contained in the intersection of:

- the auxiliary variety W, of degree at most Δ;
- the s frozen cubic equations, contributing a Bezout factor 3^s;
- the slicing equations, contributing the factor $\prod_{i=1}^{\rho} M_i(s^2)$.

Thus every zero-dimensional common-neighborhood component has degree, and hence at most this many rational points, bounded by

$$D := 3^s \Delta \prod_{i=1}^{\rho} M_i(s^2).$$

The positive-dimensional case is handled by exactly the probabilistic dimension estimate in [2, Lemma 23]. More concretely, Bukh's estimates depend on the auxiliary variety only through its dimension, pure-dimensionality, and the Bezout degree bound for the intersections under consideration. Replacing Bukh's bound 3^Z for the degree of his auxiliary variety by the present bound Δ replaces the old common-neighborhood degree bound $3^{s+Z}\prod_i M_i(s^2)$ by D, and no other part of the union bound changes.

Consequently, with positive probability the resulting graph has no positive-dimensional common-neighborhood component for any fixed s-tuple of vertices. For such a graph, every common neighborhood of s vertices has size at most D. Thus if

$$t > D = 3^s\Delta\prod_{i=1}^{\rho} M_i(s^2),$$

the graph is $K_{s,t}$-free. In Bukh's original same-degree setting, W is cut out by Z cubic equations and $\Delta \le 3^Z$, giving the displayed factor 3^{s+Z} in [2, Lemma 23]; the proof uses only $\deg W$, so the same argument gives the stated form.

Finally, [2, Lemma 24] gives

$$\prod_{i=1}^{\rho} M_i(s^2) = s^{O(s^{2/3})}$$

for the usual choice $\rho = O(s^{2/3})$.

The hypothesis gives the required varieties over all sufficiently large prime fields. By Bertrand's postulate, for every sufficiently large real x there is a prime q with $x \le q \le 2x$. Taking $x \asymp_s n^{1/s}$ gives a constructed graph on $N \asymp_s n$ vertices with $\Omega_s(n^{2-1/s})$ edges. If $N < n$, add isolated vertices. If $N > n$, take an induced n-vertex subgraph with at least the average number of edges. In either case one obtains the stated lower bound for every sufficiently large n. □

7. The main theorem

Theorem 7.1. *Fix $\varepsilon > 0$. For all sufficiently large s and all sufficiently large n,*

$$\mathrm{ex}(n, K_{s,t}) = \Omega_s(n^{2-1/s})$$

whenever

$$t > 3^s 4^{(2/3+\varepsilon)s} s^{O(s^{2/3})}.$$

Proof. Choose

$$Z = \left\lceil \left(\frac{2}{3}+\varepsilon\right) s \right\rceil$$

and choose

$$\rho = O(s^{2/3})$$

with

$$\binom{\rho+4}{3} \ge s^2.$$

By Corollary 5.3, in particular over every sufficiently large prime field, there exists a smooth complete intersection W which is s-wise cubic-independent, pure-dimensional of dimension $s+\rho$, and of degree

$$\Delta \le 4^Z.$$

Applying Proposition 6.1, Bukh's cubic random algebraic graph construction gives $K_{s,t}$-free graphs with $\Omega_s(n^{2-1/s})$ edges whenever

$$t > 3^s\Delta\prod_{i=1}^{\rho} M_i(s^2).$$

Using

$$\Delta \le 4^Z \le 4^{(2/3+\varepsilon)s+O(1)}$$

and

$$\prod_{i=1}^{\rho} M_i(s^2) = s^{O(s^{2/3})},$$

we get the displayed threshold. ☐

Corollary 7.2. *Let*

$$c_0 = 3 \cdot 4^{2/3} = 7.559526299\ldots.$$

For every fixed $c > c_0$ and all sufficiently large s,

$$\mathrm{ex}(n, K_{s,\lfloor c^s \rfloor}) = \Omega_s(n^{2-1/s})$$

for all sufficiently large n.

Proof. Choose $\varepsilon > 0$ small enough that

$$3 \cdot 4^{2/3+\varepsilon} < c.$$

Since

$$s^{O(s^{2/3})} = \exp(o(s)),$$

Theorem 7.1 implies that its threshold is below c^s for all sufficiently large s. ☐

8. Comments on sharpness of the quartic sieve

The quartic sieve is limited by five collinear points. Five points on a line form a cubic circuit. If the ambient dimension is

$$b = s + \rho + Z,$$

the family of five collinear points has dimension

$$\dim \mathrm{Gr}(1, b) + 5 = 2(b-1) + 5 = 2b + 3.$$

A quartic equation imposes 5 independent conditions on such a set. Thus using only quartics requires approximately

$$5Z > 2b + 3 = 2(s + \rho + Z) + 3,$$

or equivalently

$$Z > \frac{2}{3}s + O(s^{2/3}).$$

This is exactly the exponent used above. Hence, within the cubic-edge construction and a quartic-only auxiliary sieve, the constant

$$3 \cdot 4^{2/3}$$

is the natural endpoint.

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School of Mathematics, Shandong University, Jinan, China

Extremal Combinatorics and Probability Group, Institute for Basic Science, Daejeon, South Korea
Email address: `heng.li@sdu.edu.cn`

School of Mathematical Sciences, University of Science and Technology of China, Hefei, China
Email address: `liuxizhi@ustc.edu.cn`